INCH-POUND

MIL-DTL-55302/129E
30 January 2024
SUPERSEDING
MIL-C-55302/129D
29 April 1993

DETAIL SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT ANGLE, 90 THROUGH 240 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

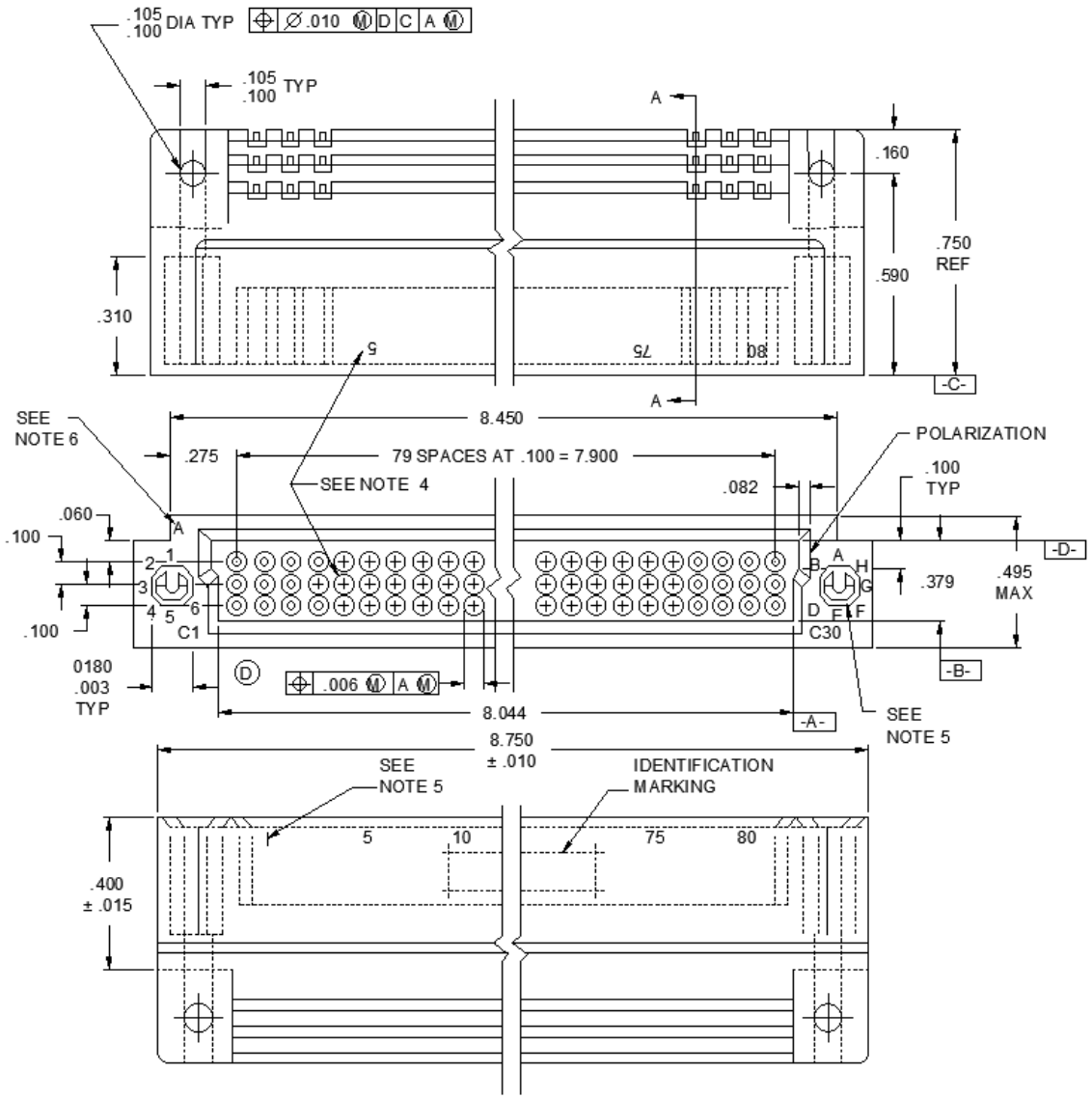
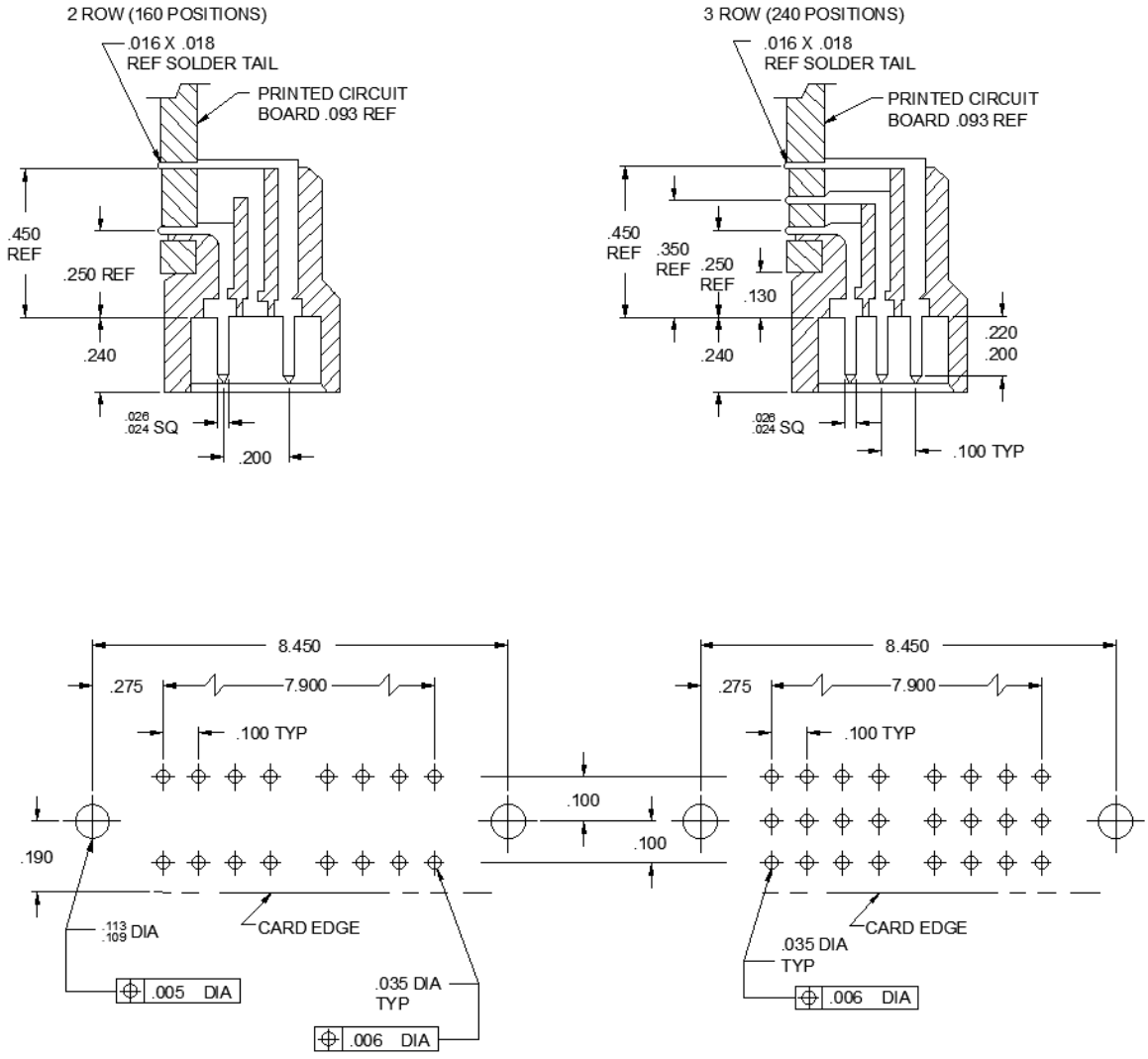



FIGURE 1. Connectors, kit plug, .100 spacing, 160 and 240 contacts.

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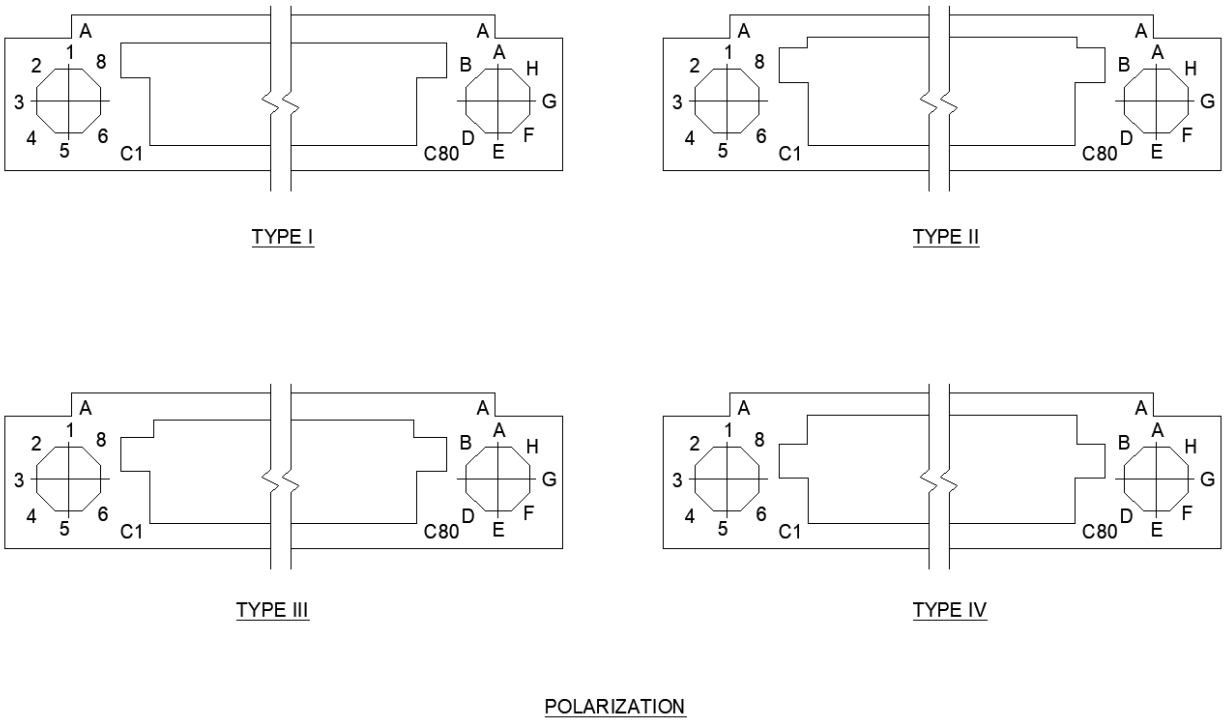



Recommended circuit layouts

FIGURE 1. Connectors, kit plug, .100 spacing, 160 and 240 contacts – Continued.



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TYPE I

TYPE II

TYPE III

TYPE IV

POLARIZATION

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 on three place decimals and +.2º on angles.
3. These connectors mate with connectors specified in MIL-DTL-55302/130.
4. Numbers indicating end cavities marked or molded both sides. Lines indicating every five cavities marked or molded on mating face, except end cavities.
5. Location indicators embossed on surface.
6. Letters indicating rows and numbers indicating end cavities marked or molded this surface.

FIGURE 1. Connectors, kit plug, .100 spacing, 160 and 240 contacts – Continued.



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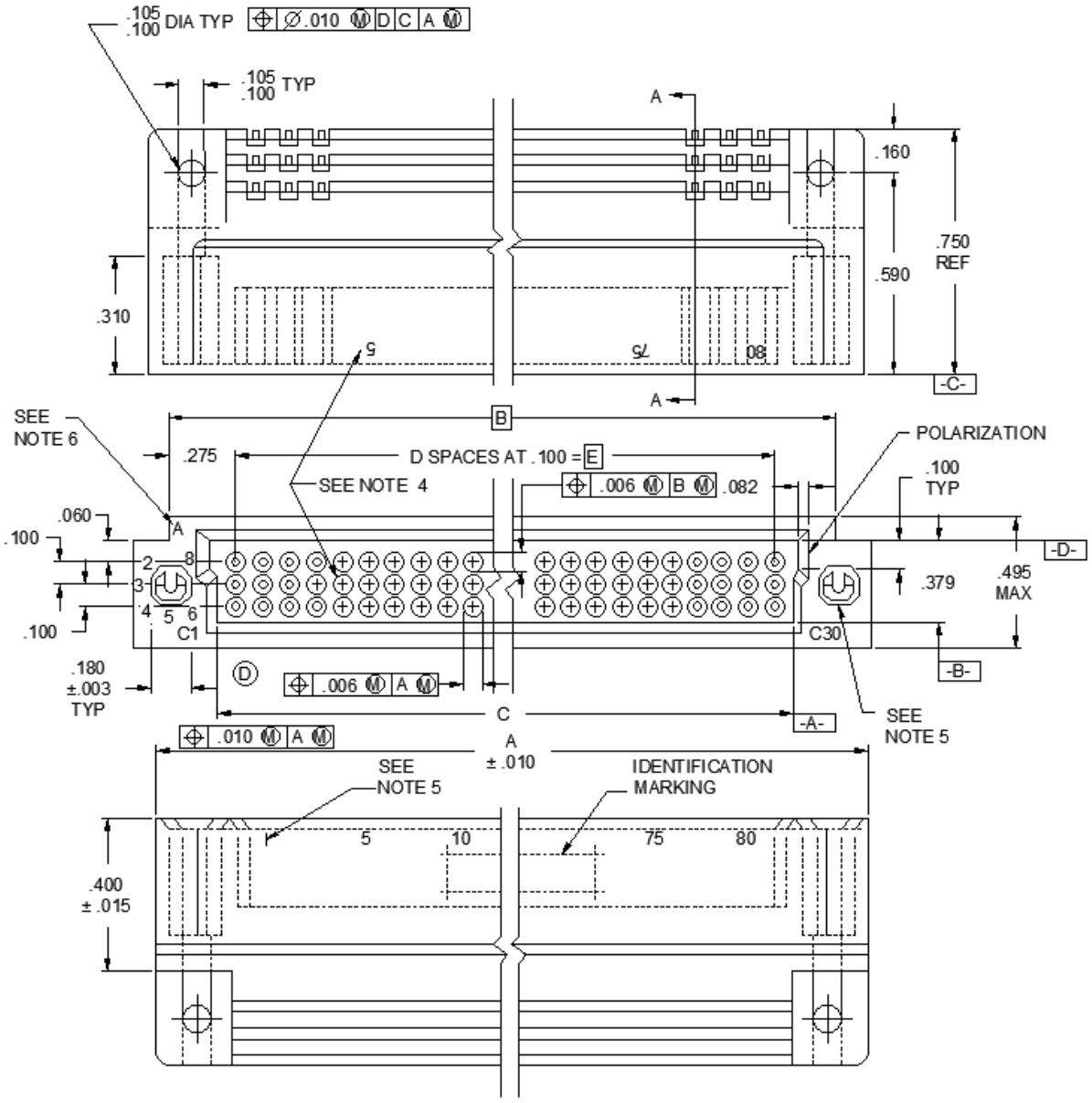
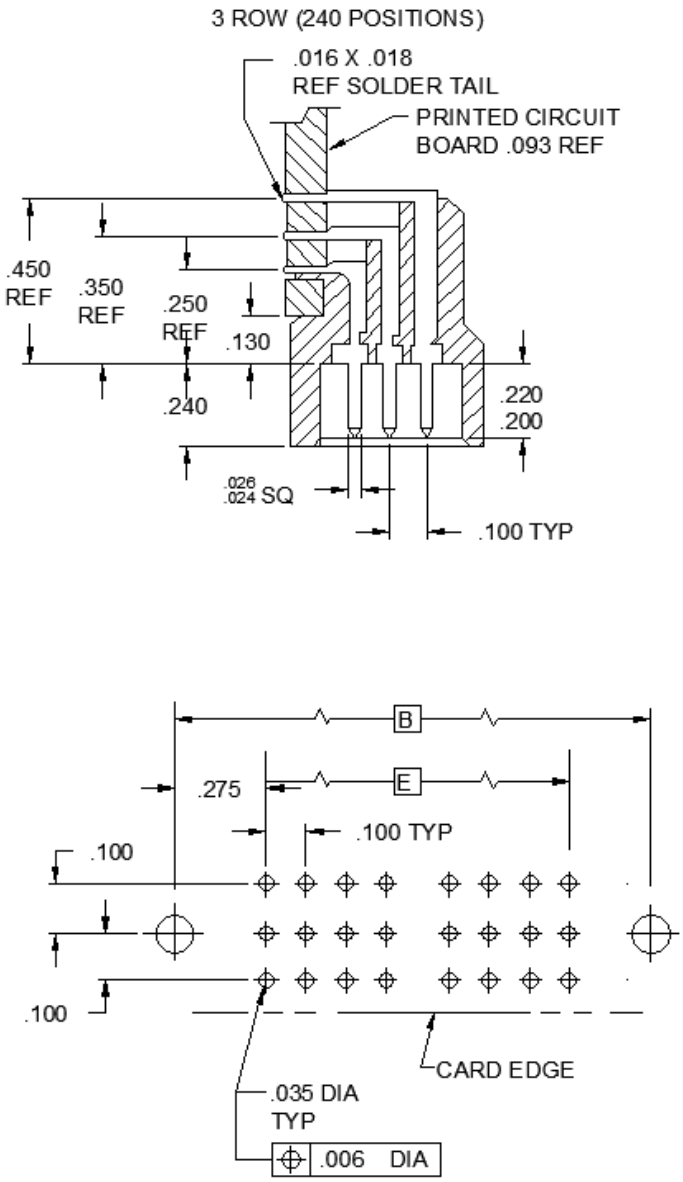



FIGURE 2. Connectors, plug, .100 spacing, 90 and 240 contact positions.



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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 on three place decimals and +.2º on angles.
3. These connectors mate with connectors specified in MIL-DTL-55302/130.
4. Numbers indicating end cavities marked or molded both sides. Lines indicating every five cavities marked or molded on mating face, except end cavities,
5. Location indicators embossed on surface.
6. Letters indicating rows and numbers indicating end cavities marked or molded this surface.

FIGURE 2. Connectors, plug, type II, .075 (1.91 mm) spacing – Continued.



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TABLE I. Dash numbers, polarization types, and contact types for 160 and 240 contact positions.

| Dash number | Number of contacts | Type polarization |
|---|---|---|
| 01 | 160 | I |
| 02 | 160 | II |
| 03 | 160 | III |
| 04 | 160 | IV |
| 05 | 240 | I |
| 06 | 240 | II |
| 07 | 240 | III |
| 08 | 240 | IV |

TABLE II. Dash numbers and dimensions (For 90 through 240 contact positions).

| Dash number | Number of contacts | Dimensions 1/ | | | | | Contact identification number |
|---|---|---|---|---|---|---|---|
| | | A | B | C | D | E | |
| 09 | 90 | 3.750 | 3.450 | 3.044 | 29 | 2.900 | 1-30 |
| 10 | 105 | 4.250 | 3.950 | 3.544 | 34 | 3.400 | 1-35 |
| 11 | 120 | 4.750 | 4.450 | 4.044 | 39 | 3.900 | 1-40 |
| 12 | 135 | 5.250 | 4.950 | 4.544 | 44 | 4.400 | 1-45 |
| 13 | 150 | 5.750 | 5.450 | 5.044 | 49 | 4.900 | 1-50 |
| 14 | 165 | 6.250 | 5.950 | 5.544 | 54 | 5.400 | 1-55 |
| 15 | 180 | 6.750 | 6.450 | 6.044 | 59 | 5.900 | 1-60 |
| 16 | 195 | 7.250 | 6.950 | 6.544 | 64 | 6.400 | 1-65 |
| 17 | 210 | 7.750 | 7.450 | 7.044 | 69 | 6.900 | 1-70 |
| 18 | 240 | 8. 750 | 8.450 | 8.044 | 79 | 7.900 | 1-80 |



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REQUIREMENTS

Design and construction:

Dimensions and configuration: See figures 1 and 2 and table II.

Material:

Insulator body: Insulator material shall be in accordance with MIL-DTL-55302 or type GLCP-30F in accordance with ASTM D5138 or MIL-M-24519.

Contact material: Contact material in accordance with MIL-DTL-55302 or Beryllium copper alloy C17400 in accordance with ASTM B768.

Plating: The contact plating for the engagement area, (.150 inch minimum length) shall be gold in accordance with MIL-DTL-55302 over nickel in accordance with MIL-DTL-55302. The contact area between the engagement area and the solder tail shall be nickel accordance MIL-DTL-55302. The contact plating for the solder tail area shall be tin-lead over nickel in accordance with MIL-DTL-55302.

Contact identification: See figures 1 and 2.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at 75ºF.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

High altitude: 200 volts rms.

Keying: See MIL-DTL-55302/31.

Part or Identifying Number (PIN): M55302/129-(and dash number from table I for 160 and 240 contact positions). M55302/129- (and dash numbers from table II for 90 through 240 contact positions).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/130
MIL-DTL-55302/31
MIL-M-24519
ASTM B768
ASTM D5138



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CONCLUDING MATERIAL

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Navy – AS, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5935-2024-214)

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